



Electronic Archiving Trust Services

CEN/TS 18170:2025 — A Cybersecurity
Perspective

Target audience: Master's in Cybersecurity Duration: 2 hours

Michel COTTIN - Archives nationales de Luxembourg (ANLux) — SCCE



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Training Plan — 120 minutes



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- **Module 0 · Introduction: history & risks of digital preservation**

T+0 → T+15 min

- **Module 1 · Why should cybersecurity professionals care about archiving?**

T+15 → T+30 min

- **Module 2 · The OAIS model: SIP / AIP / DIP**

T+30 → T+60 min

- **Module 3 · Security and trust requirements (CEN/TS 18170)**

T+60 → T+90 min

- **Module 3+ · Transferability, lifecycle & deposit to heritage archives [bonus]**

T+83 → T+110 min

- **Module 4 · Integrity chains, audit trails & reports**

T+110 → T+130 min

- **Module 5 · Summary, key points & questions**

T+130 → T+145 min (total: ~2h25)

- **Bonus · Archiving by Design — digital lifecycle (ANLux Strategy, 2024)**

Optional — ~30 additional minutes



<https://www.linkedin.com/in/michel-cottin-902a979/>



Introduction: History & Risks of Digital Preservation

T+0 → T+15 min

Case Study 1 — NASA Viking Mission (1999)



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- 1975–1982: Viking probes orbit and land on Mars. Data recorded on magnetic tapes.
- 1999: Dr. Miller requests access to the archived data — 20 years after the mission.
- The tapes are physically intact. The file formats are completely unreadable.
- No software, hardware, or documentation to interpret the bit sequences.
- What survived? Paper printouts — the most accessible long-term format.

Key principle: physical integrity ≠ logical accessibility.





1. INTEGRITY THREATS

Bit corruption, media degradation, write errors, accidental deletion

2. TECHNOLOGICAL OBSOLESCENCE

Hardware · Operating systems · Software · File formats

3. ATTACKS (cybersecurity dimension)

Ransomware · Unauthorised alteration · Unauthorised deletion / exfiltration

4. OPERATIONAL ACCIDENTS

Accidental erasure · Budget cuts / staff turnover · Loss of context (missing metadata)

Questions, comments ?

The Ship of Theseus Paradox: Preserving without Betraying



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- Traditional paper archives survive centuries passively.
- Digital archives require active and continuous technical intervention.
- The longer data must be kept, the harder it is to guarantee accessibility.

CEN/TS 18170 IS NOT a storage specification.

It is a trust service specification: processes, policies, cryptographic protections and audit mechanisms that ensure bits remain meaningful, authentic and accessible over time.

→ This is the difference between a backup and an archive.



The Athenians preserved Theseus's ship by replacing each rotting plank one by one. The philosophical question: once the last original plank was replaced, was it still the Ship of Theseus? CEN/TS 18170 is the answer to this paradox: a trust framework that ensures that despite migrations, re-signings and successive technical transformations, the document remains itself — authentic, integral, accessible.



Why Should Cybersecurity Professionals Care About Archiving?

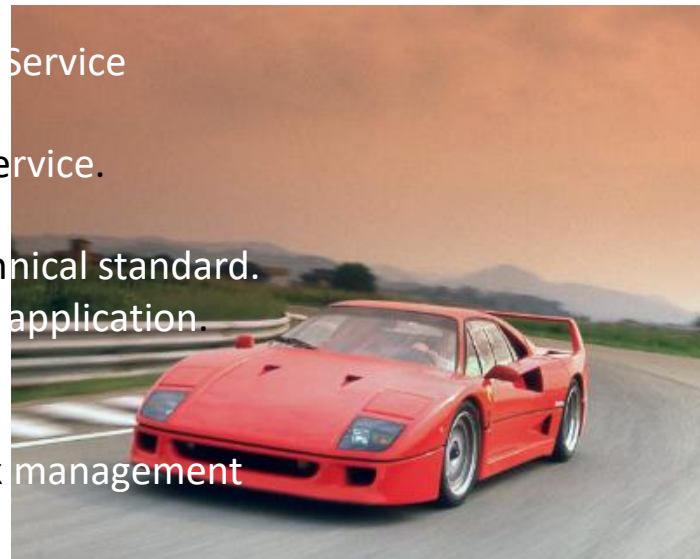
T+15 → T+30 min

The Regulatory Driver — eIDAS 2



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- Regulation (EU) 2024/1183 — eIDAS 2, in force since May 2024.
- Amends the original eIDAS Regulation (EU) 910/2014.
- Article 45j introduces the Qualified Electronic Archiving Trust Service (QEATS).
- First time long-term digital preservation is a regulated trust service.
- CEN/TC 468 (AFNOR secretariat) tasked with drafting the technical standard.
- CEN/TS 18170:2025 approved on 13 May 2025 — provisional application. This standard **is a part of the eIDAS2 implementation act.**
- NIS2 Directive (EU 2022/2555) also applies: cybersecurity risk management obligations for trust service providers.
- The lesson from previous standards (NFZ 42-013, ILNAS 106) teaches us that the requirements of qualified archiving influence different projects, including those not intended for certification.



Two Distinct Trust Services for Preservation (eIDAS)



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Type A — Art. 3(16)(e)

Preservation of electronic signatures, seals or certificates. Already in original eIDAS. Extends the cryptographic validity of signatures over time. → ETSI TS 119 511 / 512

Type B — Art. 3(48)

Electronic archiving (NEW in eIDAS 2). Guarantees durability, readability, integrity, confidentiality and proof of origin of electronic documents and data throughout the retention period. → CEN/TS 18170:2025 ← [this course](#)



Definition — CEN/TS 18170:2025 §3.1.1:

"A service ensuring the receipt, storage, retrieval and deletion of electronic data and electronic documents in order to guarantee their durability and readability and to preserve their integrity, confidentiality and proof of origin throughout the retention period."

Key actors:

- EATSP — Electronic Archiving Trust Service Provider (the operator)
- QEATSP — Qualified EATSP (following independent third-party conformity assessment)
- Subscriber — natural or legal person bound by an obligation to the EATS (§3.1.8)
- Relying party — authorised consumer of archived data

The Link with Cybersecurity (according to archivists) —Pillars



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Authenticity → **act of maintaining overtime information**, independently understandable by a designated community, and with evidence supporting its authenticity § 3.3.2 (*preservation*)

Integrity → **Fixity (hash) at ingest + periodic recomputation + cryptographic audit trail**

§9.8 (*audit AIP*), §10.1 (*AIP fixity*), §13.5 (*critical events*)

Confidentiality → **Physical/logical separation per subscriber + mutual TLS channels for AIP replication**

§10.3 (*EU-only infrastructure*), §10.4

Availability → **≥2 geographically separate sites + BCP + NIS2-compliant incident management**

§10.4, §7.12 (*BCP*), §7.10 (*incident management*)

Non-repudiation / Proof of origin → **Qualified signatures/seals on reports + timestamped events**

§14.1 (*reports*), §13.3.1 (*trusted time*)

Concept of reversal of evidence



The OAIS Model — SIP · AIP · DIP

T+30 → T+60 min

The Open Archival Information System (OAIS)

— ISO 14721



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- ISO 14721 (2003, revised 2012/2024): international reference model for long-term digital preservation.
- CEN/TS 18170 inherits and adapts its vocabulary (terms, information packages, lifecycle).

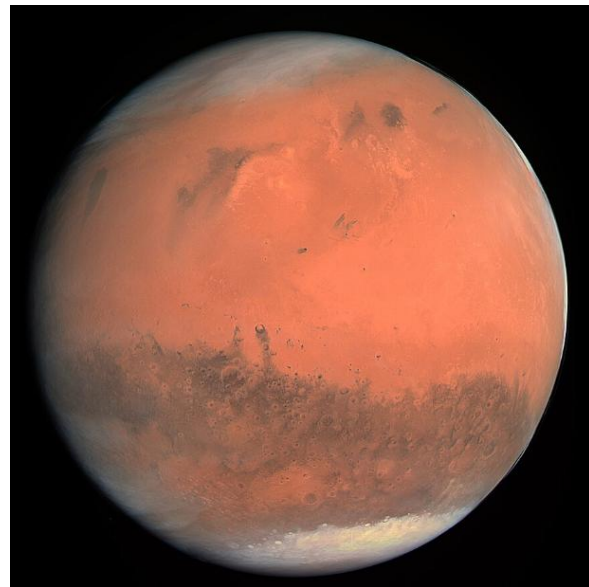
Six functional entities:

Ingest · Archival Storage · Data Management · Access · Preservation Planning · Administration

Two stakeholder roles:

- Producer (Subscriber) — submits data via the SIP
- Consumer (Relying party) — retrieves data via the DIP

Three information package types: SIP → AIP → DIP





-
- Delivered by the Subscriber to the EATS in accordance with the submission agreement.

Mandatory components (§9.3):

- Content information — the digital objects themselves (documents, datasets...)
- Preservation Description Information (PDI) — provenance, context, reference, fixity, access rights
- Packaging information — manifest describing how components are linked
- Optional: descriptive, administrative or technical metadata (e.g. MIME types)
- Example formats (§9.2): METS · BagIt (RFC 8493) · OCFL · E-ARK CSIP · XAIP
- Security: the EATSP is responsible for transfer integrity and confidentiality once protocols are respected (§9.6).



-
- Transformed from one or more SIPs. Resides in the EATS throughout the retention period.
 - Must contain: all components of the corresponding SIP(s) + transformation record (recommended: PREMIS metadata)
 - Each AIP has a unique, fixed archival object identifier within the EATS (§10.1).

Fixity (§10.1):

- Hash computed at creation and updated after each preservation action
- Recomputed periodically — results recorded as critical events

Storage (§10.4):

- ≥ 2 geographically separate sites in the EU · Subscriber separation (physical or logical)
- ≥ 1 offline or near-line copy (ransomware protection)



- Derived from one or more AIPs and delivered in response to an authorised retrieval request.
- The DIP is the "read-only view" of the archive — the Subscriber/Relying party's window onto the AIP.

Access control (§11):

- Access permissions defined by document/data scope
- A user may have the right to "search", to "retrieve" or both — independently
- Search results only display authorised information
- Format: defined in the service agreement (§11) — examples: E-ARK DIP · ETSI TS 119 512 (ASiC-E, XAIP)
- Each retrieval is traced — recorded as a non-critical event (§13.4(f)).

Preservation Description Information (PDI)



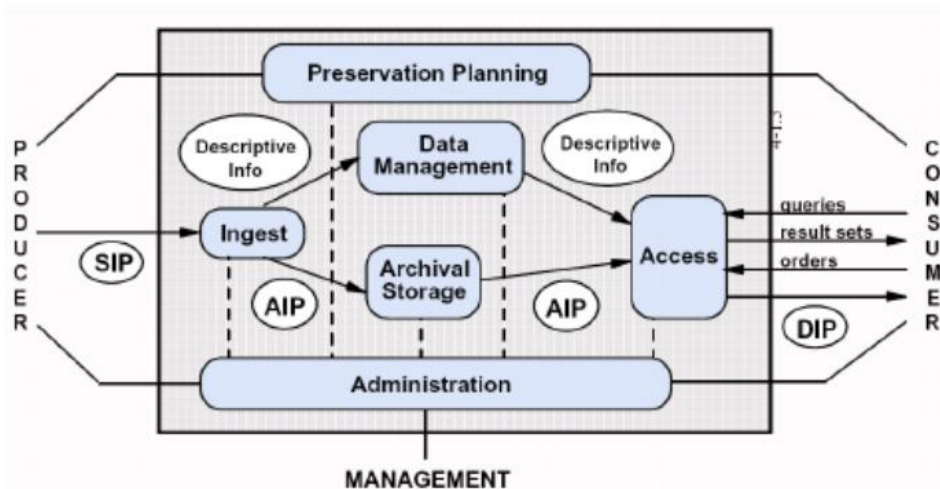
LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

PDI = what makes a digital object understandable and trustworthy over the long term.

Five mandatory categories (OAIS §3.2.4, adopted by CEN/TS 18170):

- Provenance — history and origin: who created it, how it was transferred
- Context — relationship with other objects in the collection
- Reference — identifier(s) for unambiguous identification
- Fixity — checksums (SHA-256, SHA-512...): anchor of custody chain verification
- Access rights — who can access, under what conditions

Then again... sometimes, against all odds,
it just might. ... *knowing this diagram is
probably never gonna help you at a fancy
dinner party.*





Security and Trust Requirements — CEN/TS 18170:2025

T+60 → T+90 min



Three required documents (§6.1–6.4):

- Archiving policy — rules, scope, service level, security standards
- Practice statement — how the policy is operationalised (logging, deletion, format conversion)
- Service agreement / Submission agreement — complete technical contract with each subscriber

The policy must address deletion scenarios (§6.1):

(j) At end of retention period · (k) After transfer · (l) At Subscriber's request

- Each deletion must prevent file recovery and be recorded as a critical event.
- The policy must include long-term technology planning for system maintenance and format migration.

Management and Operations — (ETSI EN 319 401)



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- §7.3 **Human resources** — EATSP team must include staff with advanced archival skills AND advanced electronic archiving skills (EQF/ESCO). Full requirements: ETSI EN 319 401 §7.4.
- §7.5 **Access control** — RBAC on AIP access.
- §7.6 **Cryptographic controls** — ETSI TS 119 312 (cryptographic suites). Monitor algorithm strength. If an algorithm becomes inadequate → update policy or create a new archiving profile. Mutual TLS between EATS components mandatory.
- §7.10 **Incident management** — ETSI EN 319 401 §7.9 + NIS2 obligations. Significant incidents reported within 24 h to the supervisory body.
- §7.13 **Cessation plan** — Formal plan: activities, roles, risks, notification timelines to all subscribers.
- §7.15 **Supply chain** — ETSI EN 319 401 §7.14. Sub-contractors handling AIPs are covered.



-
- Geographic redundancy: ≥ 2 storage sites in the EU, geographically separate at all times. Sufficiently distant to avoid common physical, natural and (geo)political risks.
 - Data separation: per subscriber — physical OR logical.
 - Offline/near-line protection: ≥ 1 copy must be near-line or offline.

Rationale: protection against ransomware affecting all online copies.

- Trusted replication channels: AIP exchanges between sites via mutual TLS.
- **EU-only constraint (§10.3): ALL storage and processing infrastructure must be located within the European Union.**
- Recovery obligation: restore within agreed timescales + notify the Subscriber.

Cryptographic Algorithm Lifecycle Management



- The EATSP continuously monitors algorithm strength in accordance with ETSI TS 119 312.

Decision process when an algorithm becomes inadequate:

- Option A: update the archiving policy for future AIPs.
- Option B: create a new archiving profile for existing AIPs + migrate signature augmentation.
- Algorithm migration events → recorded as CRITICAL events (§13.5(f)).
- Augmentation (Type A): re-timestamp existing signatures before expiry to extend their validity.

Scenario: SHA-1 deprecated. The EATSP holds 400,000 AIPs.

All fixity values and signature containers must be migrated. Full audit trail required.

This standard introduces the possibility of not using only signature augmentation

Transferability — Why It Matters



Definition — §3.3.4: "Transfer: technical and organisational process of returning archived digital objects to the Subscriber or to another EATSP authorised by the Subscriber."

- **Transferability clause: MANDATORY in every service agreement (§6.2).**

Two transfer contexts (§12.1):

- (a) Subscriber ↔ EATSP — return of digital objects at contract end or on request.
- (b) Inter-company — from one EATSP to another authorised EATSP.
- Scope of transfer: digital objects + audit trails + critical events + produced reports (§12.1).
- Transfer may be partial or total (§12.1).
- Liability remains with the EATSP throughout the service (§12.3).

Transferability (and reversibility) is a weak point, a blind spot often observed. The way to manage it must be through "by design"

The Transferability Clause — Mandatory Content



- (a) Definitions & purpose — scope of the clause (provider change, termination...)
- (b) Scope of transfer — digital objects, audit trails, metadata, tools, software, reference documents
- (c) Responsibilities of the parties — EATSP and Subscriber obligations during transfer
- (d) Return conditions — file formats, media, return timescales
- (e) Documentation & assistance — complete EATS operational documentation + transition support; completion attestation
- (f) Additional resources — compensation in case of EATSP failure to meet obligations
- (g) Confidentiality & data protection — EATSP confidentiality obligations during transfer
- (h) Miscellaneous provisions — clause validity period, termination conditions
- (i) Annexes — applicable standards, technical procedures



1. Integrity of the transferred custody chain:

Transfer must include ALL audit trail records (critical events) — not just AIPs. Integrity chain must be unbroken at destination. Receiving system must be able to verify evidence records.

2. Confidentiality during transfer:

Transfer interface must ensure integrity AND confidentiality (§12.2.3). GDPR / data protection obligations follow the data to the receiving system.

3. Anti-lock-in / data sovereignty:

Without a tested transferability clause, Subscriber data is hostage to the EATSP's survival. §12.2.2: transfer procedures must be periodically assessed in accordance with risk analysis.

4. Authorisation chain:

Functions used by the transfer process must be subject to appropriate authorisations (§12.2.3).

Transferability & Archival Lifecycle — Deposit to Heritage Archives



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- The EATS is not necessarily the final destination of digital documents.

Administrative lifecycle vs historical preservation:

- The EATS manages documents during their active and semi-active life (administrative retention period).
- Heritage archives (e.g. ANLux) take over documents of permanent historical value.
- Legal obligation for public bodies (§10.6 NOTE): before destroying, public bodies must offer documents to the competent archival authority.
- Deposit process: the DIP produced by the EATS becomes the SIP of the heritage archives.
- Heritage archives may operate their own EATS — or a separate preservation system.
- The transferability clause must explicitly cover this scenario: format, timescale, evidential documentation.

The Complete Digital Document Lifecycle

The EATS in Context



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

CREATION → CAPTURE → ACTIVE MANAGEMENT → EATS → DEPOSIT → HERITAGE ARCHIVES

- Creation: natively digital document produced by the creator.
- Capture: SIP submitted to the EATS — ingest, audit, AIP creation. [CEN/TS 18170 §9–10]
- Active management: the EATS holds the AIP, periodic integrity checks, format monitoring, access via DIP. [§10–11–13]
- End of retention period: appraisal decision. DISPOSAL

→ DESTRUCTION (no archival value): critical deletion event, deletion report produced. [§10.6]

→ TRANSFER to another EATSP (provider change): transferability clause activated. [§12]

→ DEPOSIT to heritage archives (permanent value): DIP + integrity report + PREMIS events delivered to ANLux / national archives. [§6.4.2(y), §12, §14]

- After deposit: the EATS only deletes its copies after formal acknowledgement of receipt. [§12.2.3]



Integrity Chains, Audit Trails & Reports

T+110 → T+130 min

Event Taxonomy — Critical vs Non-Critical



NON-CRITICAL EVENTS — no critical impact on integrity, readability or access rules.

May be stored in a database or log files.

Examples: successful integrity checks, login/logout, metadata access (§13.4).

CRITICAL EVENTS — critical for integrity, readability or access rules.

Require additional integrity protection (digital signature or hash chain).

Together they constitute the AUDIT TRAIL (§13.5).

Absolute rule (§13.5): if a critical event cannot be recorded, the associated service MUST NOT be provided.

Example: if the deletion event cannot be logged, deletion cannot take place.

ACID transaction https://fr.wikipedia.org/wiki/Propri%C3%A9t%C3%A9s_ACID

Mandatory Critical Events



- (a) SIP ingest — hash of each digital object recorded at ingest.
- (b) Format conversion — hash before AND after conversion; old format, new format, tool recorded.
- (c) Metadata modification — descriptive, technical, administrative.
- (d) Transfer — to the Subscriber or any other authorised destination.
- (e) Integrity anomaly — any failure of an integrity or completeness check.
- (f) Algorithm update — change of the cryptographic algorithm used.
- (g–i) User management — creation, modification, deactivation; credential change; privilege modification.
- (j) Any other operation that may affect integrity.



Exercice Critical Events



Action

1. A user logs into the system
2. SIP ingest — hash of each digital object recorded
3. A user browses an AIP's metadata
4. Format conversion from PDF to PDF/A
5. Successful periodic integrity check
6. AIP deletion at the subscriber's request
7. User access rights modified
8. Authorised DIP download
9. Cryptographic algorithm updated across all AIPs

Exercice Critical Events (solutions)



Action

- | | |
|--|-----------------|
| 1. A user logs into the system | Non critical |
| 2. SIP ingest — hash of each digital object recorded | Critical |
| 3. A user browses an AIP's metadata | Non critical |
| 4. Format conversion from PDF to PDF/A | Critical |
| 5. Successful periodic integrity check | Non critical |
| 6. AIP deletion at the subscriber's request | Critical |
| 7. User access rights modified | Critical |
| 8. Authorised DIP download | Non critical |
| 9. Cryptographic algorithm updated across all AIPs | Critical |

Integrity Protection — Approach A: Digital Signatures



- Signature / seal / timestamp / evidence record applied to critical event records.
- Directly: sign the AIP or the audit trail structure.
- Indirectly: via an aggregation method (e.g. Merkle tree with evidence records — RFC 4998 / RFC 6283).
- Augmentation techniques (re-timestamping): maintain evidential validity as algorithms age.
- Merkle tree advantage: protect thousands of events in batch with a single timestamp.
- Completeness: achieved by chaining signatures, or by strict periodicity of timestamp creation.
- An external preservation service (Type A TSP) may be used to maintain long-term verifiability.

Integrity Protection — Approach B: Integrity Chains



- Chaining principle: the hash of each item depends on the hash of all preceding items.
- Implemented by including the previous block's hash in the current block.
- Retroactive falsification becomes computationally infeasible as the chain grows.
- Critical event records are chained directly (within AIPs) or indirectly (via Merkle tree).
- Subscriber separation: each audit trail chain is independent per subscriber.
- Additional protection required (§13.6.3): at least one mechanism preventing undetectable alteration.

Options: asymmetric timestamps · consensus algorithm · external ledger service.





The EATS must produce integrity reports AUTOMATICALLY.

Production process:

Recompute the digital object's hash → compare with hash in the audit trail → if match: issue report

If mismatch: TSP incident management clause → notify Subscriber → DO NOT issue a positive report

- Mandatory report fields (§14.2): EATS identity · recipient identification · timestamp · AIP identifier(s) · operation type · date · hash value

Signature requirement:

- QEATSP (Qualified): QUALIFIED electronic signature or seal of the EATSP
- Non-qualified EATSP: ADVANCED electronic signature or seal
- Report covers: from the start of the retention period to the moment of retrieval.



Summary, Key Points & Questions

T+130 → T+145 min

Exercise “the disaster scenario”



An EATSP stores 200,000 AIPs across two European datacenters, both permanently online. On a Sunday at 3am, a ransomware attack simultaneously encrypts both sites. The service is contractually bound to restore within 72 hours. You must prepare a report to the direction.

1, Which requirement of CEN/TS 18170 was not met?

2, Which security pillar is impacted first?

3, What must the EATSP do within 24 hours under

4, Can the EATSP issue integrity reports to subscribers during the outage?

5, Bonus — *What does this scenario have in common with the NASA Viking mission from Module 0?*

Exercice “the disaster scenario”



An EATSP stores 200,000 AIPs across two European datacenters, both permanently online. On a Sunday at 3am, a ransomware attack simultaneously encrypts both sites. The service is contractually bound to restore within 72 hours.

- *Which requirement of CEN/TS 18170 was not met?* → §10.4: at least one copy must be near-line or offline at all times. The rationale is explicit in the standard: protection against ransomware affecting all online copies simultaneously.
- *Which security pillar is impacted first?* → Availability — but integrity and confidentiality may also be at risk depending on whether the attacker exfiltrated data before encrypting.
- *What must the EATSP do within 24 hours under §7.10?* → Report the significant incident to the supervisory body — this is a direct NIS2 obligation applying to all trust service providers.
- *Can the EATSP issue integrity reports to subscribers during the outage?* → No. §14 requires the hash to be recomputed and matched against the audit trail before any report is issued. No valid AIP access means no valid report.
- **Bonus** — *What does this scenario have in common with the NASA Viking mission from Module 0?*
→ In both cases the data exists but is inaccessible. Physical integrity does not guarantee logical accessibility — the core lesson of the course stated from the very first slide

11 Key Points to Remember



1. eIDAS 2 creates the QEATS — a new regulated trust service governed by CEN/TS 18170:2025.
2. Electronic archiving ≠ backup. It is a trust service covering integrity, confidentiality, availability, non-repudiation over decades.
3. OAIS model (ISO 14721): SIP (submission) → AIP (archiving) → DIP (dissemination).
4. Each package is a secured container carrying the PDI (provenance, fixity, access rights...).
5. AIPs stored in ≥2 geographically separate EU sites; ≥1 offline/near-line copy.
6. Cryptographic algorithm lifecycle: continuous monitoring + mandatory augmentation.
7. Critical events = audit trail. Protected by signatures (Merkle / ERS) or hash chains.
8. Write-ahead log principle: if a critical event cannot be recorded, the operation is blocked.
9. Integrity reports: automatic, signed/sealed, covering from ingest to retrieval.
10. CEN/TS 18170 is a Technical Specification (provisional). Conversion to a European EN expected after 2 years of review.
11. Issues for existing tools



1. Integrity chain vs blockchain

§13.6.3 allows the use of external ledger services to anchor audit trail records. Compare: re-timestamping with ERS RFC 4998 vs blockchain anchoring. Comparison points: cost, portability of verification, third-party dependency.

2. Ransomware scenario

Design an EATSP storage architecture resistant to ransomware affecting the primary data centre, while respecting agreed recovery timescales. Which storage tiers and EU sites? What is the RTO impact of an offline copy?

3. Large-scale algorithm migration

A major hash function used for the fixity of all AIPs is compromised. 400,000 AIPs. Describe the migration steps without creating an integrity uncertainty window. Which critical events must be recorded?

Further Reading & Key References



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

-
- CEN/TS 18170:2025 — Functional requirements for electronic archiving services (this course)
 - ISO 14721:2012/2024 — OAIS Reference Model (conceptual framework for digital preservation)
 - ETSI EN 319 401 — General policy requirements for trust service providers
 - ETSI TS 119 512 — Protocols for TSPs (long-term data preservation)
 - ETSI TS 119 312 — Cryptographic suites for TSPs
 - RFC 4998 / RFC 6283 — Evidence Record Syntax (ERS / XML-ERS)
 - RFC 3161 / RFC 5816 — Time-Stamp Protocol (TSP)
 - E-ARK CSIP / AIP / DIP — Common Specification for Information Packages (EU public sector)
 - ISO 16363 — Audit and certification of trustworthy digital repositories
 - EU Directive 2022/2555 (NIS2) — Cybersecurity of critical infrastructure incl. TSPs



Archiving by Design — Rethinking the Digital Lifecycle

Inspired by the ANLux National Digital Preservation Strategy (2024)

Why Traditional Archival Logic Fails for Digital Data



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Traditional paper lifecycle assumption:

Production → Active use → End of retention → Offer to Archives → Definitive preservation OR destruction

This model assumes documents remain stable between creation and archiving.

For digital data, this assumption is FALSE:

- Storage media degrade (bit corruption, media failure)
- File formats become unreadable (NASA Viking, see Module 0)
- Software ecosystems disappear; context is lost without active documentation
- Data is updated in place — historical versions silently disappear

Key finding: digital data requires active preservation from the moment of creation, not passive waiting.

The Digital Data Lifecycle — Capture Early, Preserve Actively



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Three preservation horizons (Digital Preservation Coalition, 2nd ed., 2021):

- Short-term: access for a defined period or until technological change makes it inaccessible.
- Medium-term: continued access beyond technological changes for a defined period, but not indefinitely.
- Long-term: continued access to data, or at least to the information it contains, for an indefinite period.

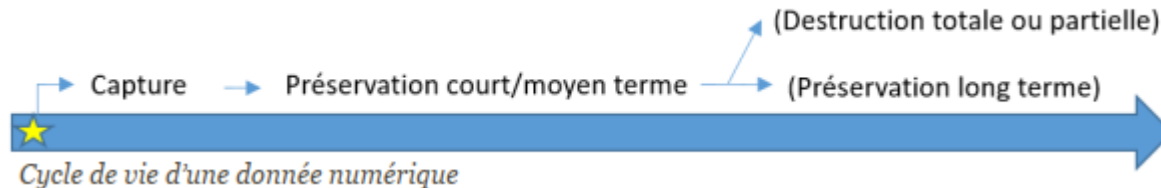
Revised digital lifecycle:

CREATION (★) → Capture → Short/medium-term preservation → [Decision point]

→ Long-term preservation (permanent value) — heritage archives

→ Destruction (no archival value) — planned, logged, irreversible

★ = the "capture moment" — as early as possible in the data creation process.



Archiving by Design — Definition & Principle



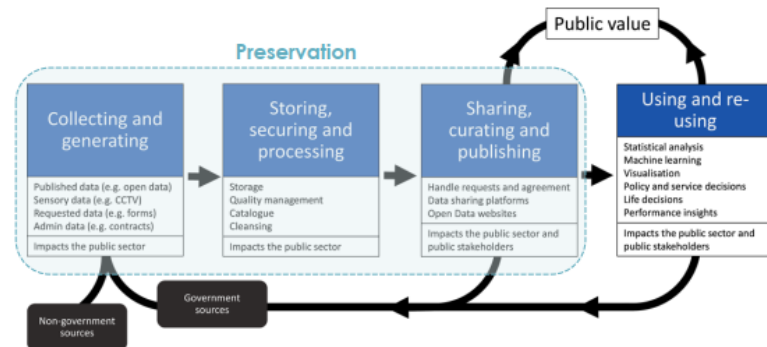
Definition (ANLux National Strategy, §3.2, 2024):

"The key takeaway is that to guarantee medium- and long-term access to information and data, their preservation must be organised from the moment of creation and integrated into a suitable system, where they will be actively monitored (archiving or preservation by design). This applies to signed electronic documents, all types of digital data, open or restricted datasets, notarial deeds, etc."

Archiving by design = preservation is not an afterthought — it is built into the data production process from day one.

Three implications:

- 1. Preservation requirements are defined BEFORE the system is built, not added after.
- 2. Metadata, provenance and context are captured at the point of production, not reconstructed later.
- 3. Retention rules and final disposition (delete / transfer / heritage archives) are encoded in the processing workflow.



D'après VAN OOIEN, UBALDI and WELBY, "A data-driven public sector: Enabling the strategic use of data for productive, inclusive and trustworthy governance", 2019, <https://doi.org/10.1787/09ab162c-en>

Five Cardinal Points of Digital Preservation (ANLux Strategy, §6)



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

Integrity & Authenticity:

Data must be accurate, complete, accessible. Provenance documented and traceable. Verifiable at any time.

Interoperability:

Data must be preserved with the ability to make it usable in third-party business applications, or remobilisable in the event of a crisis affecting essential infrastructure.

Accessibility:

Data must be available, readable and understandable to users who are entitled to consult it.

Trust & Security:

Preserve data in which today's and tomorrow's citizens will have full confidence. Preserve security and confidentiality interests in access rights management so that all stakeholders can trust the system.

The Problem of Chain of Evidence — Medienbruch



- Strategic priority: combating breaks in the administrative chain of evidence.

Medienbruch (German): break in the information chain caused by a change of medium.

Classic Medienbruch sequence:

Trusted digital source → Print → Paper signature → Scan of signed document → Storage of scanned PDF → Submission to administration

Result: the signed scan is a copy of a copy. Original digital authenticity is lost at step 1.

Archiving by design response:

- Use qualified electronic signatures (eIDAS) to sign the digital original — no printing required.
- Submit the digitally signed document directly to the EATS as a SIP — preserving the complete provenance chain.
- Goal (National Strategy, horizon 2030): eliminate Medienbruch from public sector processing workflows.

Archiving by Design — Practical Implementation



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

ANLux National Strategy — 9 preservation principles (Annex, §2):

- 1. Guarantee authenticity — hashing, fixity checks, provenance information
- 2. Preserve technically — open standards, hardware obsolescence monitoring
- 3. Preserve durable access — functional preservation, format migration, reformatting
- 4. Integrate all data types — documents, databases, logs, audio/video, code
- 5. Document, guarantee provenance, associate adequate metadata
- 6. Account for rights, permissions and restrictions — confidentiality and availability
- 7. Use recognised standards and norms — open standards for interoperability and longevity
- 8. Manage organisational resources effectively — clear responsibilities at all levels
- 9. Be transparent, measurable and auditable — regular audits and certification

The National Digital Preservation Strategy — Context



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

- Document: "National Strategy for the Preservation of Digital Data", ANLux, v1, 01/03/2024.
- Approved by: CNI (01.03.2024) · Chamber of Deputies (12.03.2024). Administrative jurisdictions and Parquet General.
- Review cycle: every 4 years minimum from publication. Minor revision in 2026.
- Strategic objective: by 2030, all digital data produced or received by Luxembourg public sector bodies are reliable, integral and sustainably managed from conception.

Three implementation phases:

- Phase 1 — Establish an enabling framework (governance, legal harmonisation, skills development)
- Phase 2 — Bring existing "essential" digital data into compliance (identify, audit, preserve)
- Phase 3 — Encourage preservation by design for all new systems (Objective 9)

Complementary with: National Cybersecurity Strategy IV · E-Government 2021–2025 · NIF · Digital Decade 2030.

Archiving by Design — Synthesis: Connecting the Threads



LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Archives nationales

The three threads of the course, brought together:

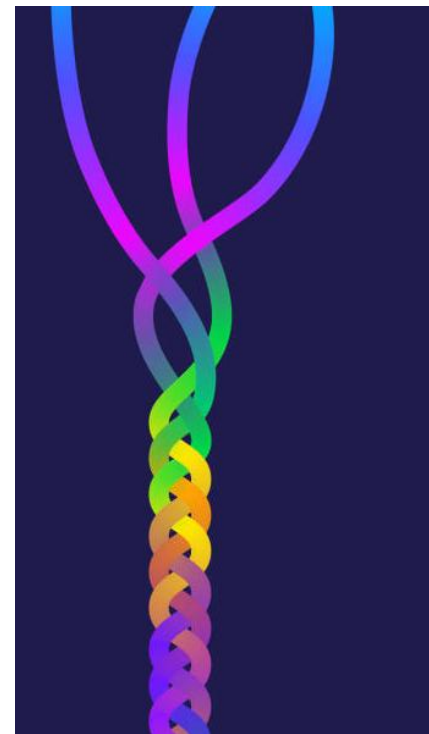
- Module 0 (History): what happens when archiving is NOT by design — NASA Viking.
- Modules 1–4 (CEN/TS 18170): the trust service framework that operationalises archiving by design in practice.
- Bonus (ANLux Strategy): the national policy framework that mandates archiving by design for all Luxembourg public bodies by 2030.

The fundamental equation:

Archiving by Design = EATS (CEN/TS 18170) + connected from creation + preservation principles + chain of evidence never broken

For cybersecurity professionals:

Archiving by design is secure-by-design applied to the data lifecycle. It is the long-term complement of security-by-design: both insist that trust properties must be built in from the start, not added as an afterthought.





Thank You

Questions & Discussion

Michel — Archives nationales de Luxembourg (ANLux / SCCE)

CEN/TS 18170:2025 (E) · ISO 14721 · eIDAS 2 (EU 2024/1183)